

1.1)

$$I = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & -3 & -4 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$O = \begin{bmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & -7 & -7 & 0 \\ 4 & 1 & 0 & 4 \end{bmatrix} //$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} * K = 2 + 1 + 1 = 4$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 1 & 1 \end{bmatrix} * K = 1 + 2 - 3 + 1 = 1$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} * K = 1 + 2 + 1 + 1 = 5$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & -3 & -4 \\ 1 & 1 & 1 \end{bmatrix} * K = 1 + 1 - 6 - 4 + 1 = -7$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} * K = 1 + 2 + 1 + 2 = 6$$

$$\begin{bmatrix} 1 & 2 & 1 \\ -3 & -4 & 1 \\ 1 & 1 & 1 \end{bmatrix} * K = 2 - 3 - 8 + 1 + 1 = -7$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 1 & 0 \end{bmatrix} * K = 1 + 2 + 1 = 4$$

$$\begin{bmatrix} 2 & 1 & 0 \\ -4 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} * K = 1 - 4 + 2 + 1 = 0$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -3 \end{bmatrix} * K = 1 + 2 + 1 + 1 = 5$$

$$\begin{bmatrix} 0 & 1 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} * K = 1 + 2 + 1 = 4$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & -3 & -4 \end{bmatrix} * K = 1 + 1 + 2 + 2 - 3 = 3$$

$$\begin{bmatrix} 1 & -3 & -4 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} * K = -3 + 1 + 2 + 1 = 1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -3 & -4 & 1 \end{bmatrix} * K = 1 + 1 + 4 + 1 - 4 = 3$$

$$\begin{bmatrix} -3 & -4 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} * K = -4 + 1 + 2 + 1 = 0$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 0 \end{bmatrix} * K = 1 + 2 + 2 + 1 = 6$$

$$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} * K = 1 + 1 + 2 = 4$$

1.2) $\text{ReLU}(x) = \max(x, 0)$

$$\text{ReLU}(O) = \begin{bmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 4 \end{bmatrix}$$

1.3)

$$w_{out} = h_{out} = \frac{4 + 2 \times 0 - 2}{2} + 1 = 2 \quad O = 2 \times 2$$

$$\max \begin{bmatrix} 4 & 5 \\ 5 & 3 \end{bmatrix} = 5$$

$$\max_{\text{pool}}(O) = \begin{bmatrix} 5 & 6 \\ 4 & 4 \end{bmatrix} //$$

$$\max \begin{bmatrix} 6 & 4 \\ 3 & 6 \end{bmatrix} = 6$$

$$\max \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} = 4$$

$$\max \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} = 4$$

1.4)

$$O_{\text{flattened}} = [5 \ 6 \ 4 \ 4]^T //$$

1.5)

$$x = \begin{bmatrix} 5 \\ 6 \\ 4 \\ 4 \end{bmatrix} \quad W = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

$$Wx = \begin{bmatrix} 1 \times 5 + 2 \times 6 + 3 \times 4 + 4 \times 4 \\ 5 \times 5 + 6 \times 6 + 7 \times 4 + 8 \times 4 \end{bmatrix} = \begin{bmatrix} 45 \\ 121 \end{bmatrix} //$$

1.6)

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

$$\text{Softmax}(x_1) = \frac{e^{45}}{e^{45} + e^{121}} \approx 0$$

$$\text{Softmax}(x_2) = \frac{e^{121}}{e^{45} + e^{121}} \approx 1$$

$$O = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad 0+1=1 \therefore \text{2nd Class}$$